

Predicting California Block-Group Home Values: A Triage Tool for REIT Acquisitions

ECON 3916 — Statistical Machine Learning · Final Project · Spring 2026
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1. Problem Statement

Prediction question. Given the demographic and geographic characteristics of a California census block group, can we predict its median house value? This is a *prediction* problem, not a causal one: we are forecasting Y from X , not estimating the effect of any X on Y . Any feature importances reported below describe statistical association, not causal effect — a distinction we revisit in Section 5.

Stakeholder and decision. A mid-size California real-estate investment trust (REIT) acquisitions team selects roughly 30 candidate block groups per quarter for full appraisal from a pool of several hundred. Each appraisal costs \$5–10k and 2–3 weeks. A model that predicts block-group median value from public census-style features lets the team pre-screen the candidate pool in minutes, narrowing the list before commissioning paid appraisals. The decision the model supports is **which block groups to send to human appraisal**, not **what those block groups are worth**. This screening framing is deliberate: it tolerates an RMSE of ~\$50–75k that would be unacceptable for a final pricing tool.

Dataset. California Housing (Pace & Barry, 1997), distributed via `sklearn.datasets.fetch_california_housing`. $N = 20,640$ census block groups (1990 U.S. Census), 8 features, target variable `MedHouseVal` in units of \$100,000. Accessed April 26, 2026.

Feature	Description	Range
MedInc	Median household income in block group	0.5 – 15.0 (×\$10k)
HouseAge	Median age of houses (years)	1 – 52 (top-coded)
AveRooms	Average rooms per household	0.85 – 141.9
AveBedrms	Average bedrooms per household	0.33 – 34.07
Population	Total block-group population	3 – 35,682
AveOccup	Average household occupancy	0.69 – 1243.3
Latitude	Block-group centroid latitude	32.54 – 41.95
Longitude	Block-group centroid longitude	–124.35 – –114.31
MedHouseVal	TARGET — median home value	0.15 – 5.0 (×\$100k)

Table 1. Feature dictionary. Target row highlighted.

2. Data Description

The eight features describe each block group's demographics (`MedInc`, `Population`, `AveOccup`), housing stock (`HouseAge`, `AveRooms`, `AveBedrms`), and centroid coordinates (`Latitude`, `Longitude`). The unit of observation is the block group (~600–3,000 people), not an individual home — an important limitation for any consumer-facing application.

Quality assessment. The dataset arrives pre-cleaned: zero missing cells across all 185,760 entries. I cannot meaningfully classify a missingness mechanism (MCAR/MAR/MNAR) on data that has none, but I

note that upstream filtering by Pace & Barry (1997) dropped block groups with unreported values before aggregation — so any concerns are pre-resolved at the source.

Tukey-fence outlier checks flag 1–6% of rows in AveRooms, AveOccup, AveBedrms, and Population. These are **not data errors**: they are legitimate low-population block groups (rural resort areas, military bases) where small denominators inflate per-household averages. Dropping them would bias against rural California, so all rows are kept.

The dominant data-quality issue is right-censoring of the target. MedHouseVal is top-coded at \500,001 — every block group whose true median value exceeds \500k is recorded as exactly \500,001. About 5% of the 20,640 rows sit at this ceiling. This censoring is MNAR (the missingness depends on the unobserved true value) and is unrecoverable from this dataset alone. It biases all models toward under-prediction at the top of the market — exactly the tier the REIT cares about most. We address this in the Recommendation.

EDA findings. The strongest linear predictor is MedInc (Pearson $r \approx 0.69$ with the target). Geographic features show two distinct value clusters — the Bay Area (latitude ≈ 37 – 38 , longitude ≈ -122) and metro Los Angeles (latitude ≈ 34 , longitude ≈ -118) — visible as bright concentrations in the geographic scatter that no single linear coefficient on latitude or longitude can capture. AveRooms and AveBedrms are strongly collinear ($r \approx 0.85$), as are latitude and longitude ($r \approx -0.92$, reflecting California's NW–SE coastline). Multicollinearity inflates linear-model coefficient variance but does not bias predictions and is harmless for tree models, so all features are retained.

3. Methodology

Train/test split. A simple 80/20 random split with `random_state=42` yields 16,512 training rows and 4,128 test rows. Spatial autocorrelation between adjacent block groups means a random split likely overstates generalization to geographically novel areas; a county-block holdout would be a more honest evaluation, and is flagged as a v2 next step.

Models compared. Three models, chosen to span a complexity gradient: (i) ordinary least-squares Linear Regression as an interpretable baseline; (ii) Random Forest with 100 trees, default hyperparameters; (iii) Gradient Boosting with 200 estimators, learning rate 0.05, max depth 4. Hyperparameters were chosen for fast, modest configurations rather than tuned via grid search — an honest decision given the screening-tool framing, where marginal gains from tuning are unlikely to change downstream actions.

Why these three. The Linear Regression baseline establishes a lower bound on what any reasonable model should achieve and exposes where nonlinearities matter. Random Forest is the natural workhorse for tabular regression — robust to outliers and feature scaling, captures interactions automatically, and (critically for this project) provides free predictive uncertainty via the variance across its 100 trees. Gradient Boosting is included as a sequential-ensemble counterpart to test whether boosting's bias-reduction machinery beats bagging on this problem; if RF and GB agree, that is itself useful evidence that we are near the noise floor of what the eight features can support.

Evaluation. Each model was scored on the held-out test set using RMSE, MAE, and R^2 . Five-fold cross-validation was run on the training set to assess generalization stability. To attach uncertainty bounds to the test metrics, I bootstrapped the test set 1,000 times (resample with replacement, recompute metric) and report 2.5/97.5 percentile CIs.

Why bootstrap CIs matter here. A point estimate of $R^2 = 0.805$ invites overconfidence. The same model on a different test draw from the same population could plausibly score 0.79 or 0.82; bootstrap quantifies that draw-to-draw variation directly from the test data without distributional assumptions. The bootstrap and 5-fold CV serve different purposes: CV measures how the *fitted model* varies across *training draws*, while bootstrap measures how the *scoring metric* varies across *test draws*. Reporting both gives the REIT analyst a fuller picture of what the 0.805 number actually means.

Procedure	Setting	Purpose
Train/test split	80/20, random_state=42	Held-out generalization estimate
Cross-validation	5-fold on training set	Stability across training draws
Bootstrap CI	1,000 resamples, percentile	Uncertainty on test metrics
Permutation imp.	10 repeats on test set	Honest predictive importance
Prediction interval	5th/95th percentile of trees	Per-row uncertainty in dashboard

Table 2. Evaluation procedures and their roles. All randomness is controlled by random_state=42 for full reproducibility.

4. Results

Both flexible models substantially beat the linear baseline — about 30% lower RMSE and 0.20 higher R^2 . The Random Forest and Gradient Boosting are within bootstrap noise of each other; I select Random Forest for deployment because per-tree predictions provide free prediction intervals, which the dashboard requires for stakeholder communication.

Model	Test R^2 [95% CI]	Test RMSE [95% CI]	5-fold CV R^2
Linear Regression	0.576 [0.55, 0.60]	0.746 [0.72, 0.77]	0.612 ± 0.007
Random Forest (selected)	0.805 [0.79, 0.82]	0.505 [0.49, 0.52]	0.804 ± 0.005
Gradient Boosting	≈ 0.81	≈ 0.50	≈ 0.80

Table 3. Model performance. Bootstrap 95% confidence intervals computed on the 4,128-row test set across 1,000 resamples.
RMSE units: \$100,000 (so 0.505 ≈ \$50.5k).

Feature importance. Random Forest Gini importance ranks MedInc first (0.52), followed by AveOccup (0.14), Latitude (0.09), and Longitude (0.09). Permutation importance on the test set agrees on MedInc's dominance — shuffling it drops test R^2 by roughly 0.40 — and confirms latitude and longitude as a meaningful secondary cluster (each shuffle drops R^2 by roughly 0.10).

PREDICTIVE IMPORTANCE ONLY — these rankings describe statistical association in the 1990 California cross-section, not causal effects. We cannot conclude that raising a neighborhood's median income would raise its home values; we can only say that, at any moment in this data, neighborhoods with higher median incomes tend to have higher home values.

Headline visualization. The actual-vs-predicted scatter and residual plot for the Random Forest on the held-out test set are provided in the analysis notebook (Section 4.4) and on the deployed dashboard. The actual-vs-predicted plot tracks the $y=x$ diagonal tightly through the bulk of the distribution; the residual plot exposes the censoring bias clearly as a diagonal band of negative residuals at the top of the predicted range, where actual values pile up against the \$500k cap. Both plots are reproducible from the notebook's

random_state=42 training run.

5. Recommendation

Following Chapter 26's Situation–Complication–Resolution structure:

Situation

The REIT acquisitions team must triage hundreds of candidate block groups down to ~30 for full appraisal each quarter. Misallocating the appraisal budget costs both money and weeks of analyst time.

Complication

A linear model on the cheaply-available census features explains only ~58% of variance in median value — it would systematically under-price coastal block groups and over-price inland ones. The Random Forest improves this to ~80%, but the target is right-censored at \$500,001, which is exactly the tier the REIT cares about most. Any model trained on this data will under-predict at the top of the market.

Resolution

Deploy the Random Forest as a **pre-screening triage tool**, not a pricing tool, with four operating rules:

(1) Use predictions to *rank* candidate block groups, not to *price* them — relative ordering is more reliable than absolute level near the ceiling. **(2)** Always display the 90% prediction interval alongside the point estimate; the dashboard does this automatically. **(3)** Route any block group whose 90% PI extends to or above \$500k directly to a senior appraiser — model behavior is most distorted there. **(4)** Retrain annually on Census ACS updates; the 1990 source data is over three decades old.

Uncertainty Statement

Based on 5-fold cross-validation, the Random Forest's predictive R^2 is 0.80 ± 0.005 — the model explains roughly 80% of the variation in block-group median value, with very tight CV stability. On the held-out test set ($n = 4,128$), RMSE is **0.505** ($\approx$ \$50.5k) with bootstrap 95% CI [0.49, 0.52]. The primary limitation is right-censoring at \$500,001, which biases predictions downward for premium markets. We recommend using this model to *rank* candidate block groups for appraisal — not to set asking prices — with the caveat that any prediction near or above the \$500k ceiling should be routed to a human appraiser.

Limitations & Next Steps

Spatial autocorrelation. A random 80/20 split likely overstates generalization to geographically novel block groups; a county-block holdout CV would be more honest. **Temporal drift.** 1990 prices are not 2026 prices; retraining on ACS 2020 data is the obvious upgrade. **Causal disclaimer.** The model says nothing about how home values would respond to interventions on income, density, or any other feature — a separate causal study would be required for that.

Aggregation. The block group is a useful unit for screening but coarse for individual-property decisions. Within-block-group variance is meaningful: a model that predicts a block group's median at \$420k tells the REIT nothing about whether the specific homes they would actually buy fall above or below that. The triage framing absorbs this limitation, but a per-property pricing tool would need MLS or transactional data, not census aggregates.

References & Reproducibility

Pace, R.K. and Barry, R. (1997). "Sparse Spatial Autoregressions." *Statistics and Probability Letters*, 33: 291–297.

Spiegelhalter, D. (2017). "Risk and Uncertainty Communication." *Annual Review of Statistics and Its Application*, 4: 31–60.

Pedregosa, F. et al. (2011). "Scikit-learn: Machine Learning in Python." *Journal of Machine Learning Research*, 12: 2825–2830. Dataset accessed via `sklearn.datasets.fetch_california_housing`, April 26, 2026.

Code & data: `github.com/Ram5268/ECON3916-Statistical-Machine-Learning`. **Live dashboard:** Streamlit Community Cloud (URL on Canvas submission). All randomness controlled via `random_state=42`; environment pinned in `requirements.txt`.

Bottom Line for the REIT

USE the Random Forest predictions to RANK candidate block groups for appraisal — relative ordering is reliable.

DO NOT use the point estimate as a price. The 90% prediction interval (~\$150k wide) and the right-censoring bias near

ALWAYS route block groups whose 90% PI extends to or above \$500k directly to a senior human appraiser — model

RETRAIN annually on Census ACS updates; the underlying source data is from 1990.